Learning goals: Solving matrix equations in R, deleting columns in dataframes and lists, merging dataframes, reading excel files, sets and operations on sets, if-else statement structure (conditionals), loops, writing functions, timing your R code

EXERCISES

This lab will use the data used in the study described in "Survival of patients with primary brain tumors: Comparison of two statistical approaches", PLoS One, 11, e0148733 (2016).

(1) **Solving matrix equations and review of matrix operations.**
   (i) Form a matrix `amat` of dimensions 3×4 containing numbers from 10 to 120 in steps of 10. Print this matrix. By default the `byrow` option in `matrix()` function is `TRUE`, so repeat the above with `FALSE` and form another matrix called `amat2`. Print this matrix. What is the relation between the 2 matrices?
   (ii) Assign the row names to `amat` as `R1, R2 and R3` and column names as `C1, C2, C3, C4`. Print the matrix again to see that the assignments have been made.
   (iii) Form a 4x4 matrix `A` with the following elements: 2,5,7,3,1,8,9,10,1,12,5,10,4,17,15,11 and another matrix `B` of dimensions 4x4 with the elements: 12,5,3,17,1,18,9,10,1,12,5,10,4,15,15,4. Print both the matrices. Perform an element -wise multiplication of the two matrices and print the result. Perform a matrix-matrix multiplication of the 2 matrices and print the result.
   (iv) Define a vector `X` with elements 5,6,8,9 and vector `Y` with elements 8,10,12,5. Obtain the outer product of the two vectors and print the result. Also obtain the inner product of the two vectors and print the result.
   (v) Form a diagonal matrix using the entries of the vector `X` above.
   (vi) Print the diagonal elements of `A`.
   (vii) Create an identity matrix of dimensions 6x6 in one line.
   (viii) Create a 3x3 matrix `A` using the elements 3,4,-2,4,-5,1,10,-6,5 with default options. Print `A`.
   (ix) Create a 3x1 matrix `B` with elements 5,-3,13. Print `B`.
   (x) Find the unknown vector `X` of the equation `AX=B` using the command `X=solve(A,B)`. Print `X` to see the results. What type of object is `X`?
   (xi) Find the inverse of matrix `A` by using the command `Ainv = solve(A)`. Print the inverse. Check that this is indeed the inverse by calculating the matrix product of `Ainv` and `A` matrix and print the result. Are you getting an identity matrix?
   (xii) Find the eigenfunctions and eigenvalues of `A` by using the command `eigen(A)`. Save this to an object called `results`. What type of object is `results`? Perform a matrix-vector multiplication operation of `A` and one of the eigenvectors (say the second eigenvector). What is the result? Do you understand this result? Explain.

(2) **Removing a column from a data frame.** Read in the file "BrainCancer.csv" and store the data as a dataframe.
   (i) Create a new column of data that calculates the following: In a given row, the entry from 'GTV' column should be squared and added to the 'time' column entry in that row. Add this column to the existing data frame after giving it a column name.
   (ii) Print the row and column names of the modified data.
   (iii) Change the row names of the data such that each row has 'Row-' followed by the row number. Use the `paste` function for this.
   (iv) Remove the column 'ki' by assigning `NULL` value to this column. Print the modified data to make sure that the 'ki' column has indeed been removed.

(3) **Reading excel files.**

(i) Install the package called `readxl` by using the command `install.packages("readxl")`.
(ii) Load the package in the current R environment by using the command `library(readxl)`.
(iii) If you have done the above correctly, your next command will load successfully to create a dataframe of the data present.
`data <- read_excel('<path_to_pone.0148733.s001.xlsx>',1)`. The 1 stands for the sheet number in the excel file.
(iv) Print the column names and dimensions of the data in the `data` data frame.

(4) **Sets and operations on sets.**
- (i) Define a vector `A` with elements a,b,c,d,e. Also define another vector `B`with elements d,e,f,g. Print the vectors.
- (ii) Perform a union operation between the two sets and print the result.
- (iii) Perform an intersection operation between the two sets and print the result.
- (iv) Perform a difference operation between the two sets and print the result.
- (v) The function `setequal()` checks for the equality of two objects. Create an object with a sequence of `A-B`, `A`$\cap$ `B` and `B-A` and use this function to check its contents with the result of the union operation above. Print the result.
- (vi) List the elements of `B` present in `A` using two different approaches.
- (vii) Print the elements of `A` present in `B`.

(5) **Practice with subsets.**
- (i) Form a vector with elements 8,10,12,7,14,16,2,4,9,19,20,3,6. Print the following filtered data from this vector: (a) values greater than 12, (b) values greater than 10 and less than 20
- (ii) Form an array `A` with the elements 2,7,29,32,41,11,15,NA,NA,55,32,NA,42,109. Form a new array from this one where the values 'NA' have been removed and the values are less than 100. Print this array.
- (iii) Assign the value 0 to the elements 'NA' in `A` and print the new array.
- (iv) Create a vector with gene names "gene-1","gene-2" ... "gene-6". Create a vector for gender with entries M,M,F,M,F,F,M.
- (v) Enter the following data as 7 vectors:

```
result1 = c(12.3, 11.5, 13.6, 15.4, 9.4, 8.1, 10.0)
result2 = c(22.1, 25.7, 32.5, 42.5, 12.6, 15.5, 17.6)
result3 = c(15.5, 13.4, 11.5, 21.7, 14.5, 16.5, 12.1)
result4 = c(14.4, 16.6, 45.0, 11.0, 9.7, 10.0, 12.5)
result5 = c(12.2, 15.5, 17.4, 19.4, 10.2, 9.8, 9.0)
result6 = c(13.3, 14.5, 21.6, 17.9, 15.6, 14.4, 12.0)
result7 = c(11.0, 10.0, 12.2, 14.3, 23.3, 19.8, 13.4)
```

- (vi) Create a dataframe with the following columns: genes, gender, result1, result2, result3, result4, result5, result6, result7. Call this data frame as `datframe`.
- (vii) Add column names to this dataframe: "GeneName", "Gender", "expt1", "expt2", "expt3", "expt4", "expt5", "expt6", "expt7".
- (viii) Create a subset of data with "expt2" values greater than 20
- (ix) Create a subset of data with only Female gender
- (x) Create a subset of data with Male gender for which "expt2" is less than 30.

(6) **If-else-if structure.**
- (i) Write an if-else-if structure that explores and prints the quadrant in which an angle belongs. For example, if you input 45^o it should print 'First quadrant'.
- (ii) Write an if-else-if structure that takes three numeric inputs and uses this structure alone to put the 3 numbers in decreasing order. Do not use any built in function for sorting purposes.

(iii) Let's say the cost of a journey ticket depends not only on the distance travelled but also on the details of the traveller. Distance-wise, the cost is a minimum of Rs. 100 for the first 100km, Rs. 1.50 for every extra km until 1000km and Rs.2 per km thereafter. On top of that, senior citizens ($>$ 60 years) get a 25% concession and children under 6 years of age get 50% concession. Write a code that takes the journey distance and the traveller's age as inputs, and prints out the ticket cost.

(7) **Writing functions.**

(i) Write a function to replace all the negative values in a vector by zeros.

(ii) Write a function to calculate the factorial of a number using the Stirling's approximation:

$$n! = n^n e^{-n}\sqrt{2\pi n}\left[1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} - \frac{571}{2488320n^4}\right] \quad (1)$$

(iii) Write a function to sum the digits of a number.